DRAWING LIST

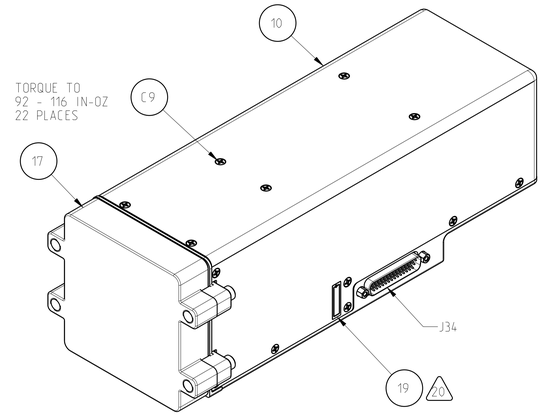
ITEM	QTY	DRAWING NUMBER	DESCRIPTION
1	1	S1101EFA2214	CAMERA ENCLOSURE HARNESS ASSEMBLY
2	3	S1101EFA2216	HALL EFFECT SENSOR BOARD ASSEMBLY
3	3	S1101EFA2221	MODIFIED CAMERA MOTOR
4	1	S1101MFA2201	CAMERA MOUNTING PLATE ASSEMBLY
5	1	S1101MFA2203	CAMERA ENCLOSURE END PANEL ASSEMBLY
6	1	S1101MFA2205	ZOOM LENS MOUNT ASSEMBLY
7	3	S1101MFA2207	ZOOM LENS MOTOR CAM ASSEMBLY
8	1	S1101MFA2210	CAMERA TOP BRACKET ASSEMBLY
9	2	S1101MFA2218	SPUR GEAR ASSEMBLY
10	1	S1101MFD2212	CAMERA ENCLOSURE COVER
11	1	S1101MFD2213	CAMERA THERMAL PAD
12	3	S1101MFD2215	SENSOR BOARD WIRE STRAIN RELIEF
13	1	S1101MFD2217	LENS ALIGNMENT RING
14	4	S1101MFP2223	MODIFIED CAMERA ENCLOSURE MOUNTING FASTENER
15	1	S1101MFA2219	MODIFIED CAMERA ASSEMBLY
16	1	S1101MFA2224	FOCUS RING SPUR GEAR
17	1	S1101MFA2225	CAMERA ENCLOSURE LENS COVER ASSEMBLY
18	1	S1101MFT2200-XX	COMBO LABEL
19	1	S1101MFT2200-04	CONNECTOR ID LABEL
20	1	S1101MFT2200-05	CONNECTOR ID LABEL

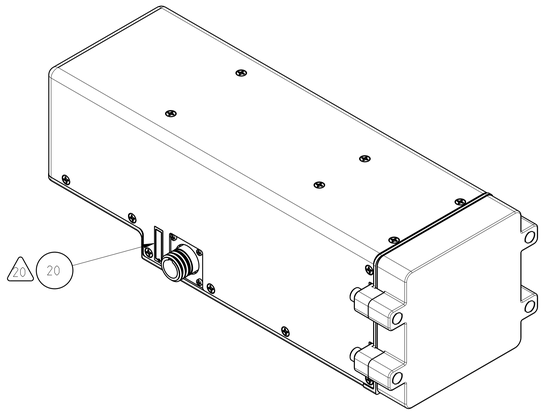
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
C1	-	REMOVED	REMOVED
C2	-	REMOVED	REMOVED
C3	4	NA0069-A030008	M3-0.5 X 8MM SOCKET HEAD CAP SCREW, A286
C4	1	1-22034	MODIFIED NAVITAR ZOOM 7000 18 - 108 MM MACRO ZOOM LENS
C5	1	809-199-8	MOUNTING FLANGE, 3/4 PERIMETER, 800 SERIES CONNECTORS, GLENAIR
C6	1	CSD-3	NUT PLATE, MIL-DTL-24308 CONNECTOR, SHELL SIZE 25 .112-40 UNJC-3B THREAD, STROCO
C7	2	JAX103-M09-F09-440	JACK SCREW, #4-40 X .312 LG. THREAD 303 STAINLESS STEEL, PASSIVATED, UNICORP INC.
C8	6	MS51045-1	.086-56 UNJC-3A X .125 HEX SOCKET SETSCREW HALF-DOG POINT, CRES, PASSIVATED
C9	22	NAS1102E04-4	.112-40 UNJC-3A X .250", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C10	3	NAS1102E06-8	.138-32 UNJC-3A X .500", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C11	2	NAS1102E06-10	.138-32 UNJC-3A X .625", 100° FLAT HEAD, OFFSET CRUCIFORM MACHINE SCREW
C12	6	MS51045-10	.112-40 UNJC-3A X .188 HEX SOCKET SETSCREW HALF-DOG POINT, CRES, PASSIVATED.
C13	6	NAS1101E00-3	SCREW, MACHINE-FLAT FILLISTER HEAD .060-80 UNJF-3A X .188 LENGTH
C14	4	NAS1101E02-5	SCREW, MACHINE-FLAT FILLISTER HEAD .086-56 UNJC-3A X .312 LENGTH
C15	4	NAS620C10L	FLAT WASHER, REDUCED OUTSIDE DIAMETER NO. 10 (.1900), .032 THICK, CRES
C16	8	NAS620C5L	FLAT WASHER, REDUCED OUTSIDE DIAMETER NO. 5 (.1250), .016 THICK, CRES
C17	A/R	222MS	LOCTITE 222MS THREAD LOCKER
C18	A/R	3145 GRAY	RTV ADHESIVE/SEALANT, GRAY, DOW CORNING
C19	A/R	THFE4-180010	HEAT SHRINK TUBING, .180 ID, SAINT-GOBAIN

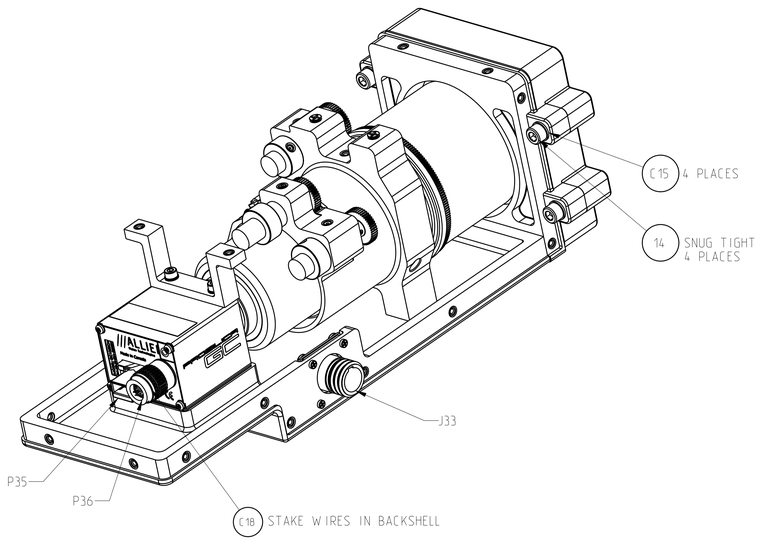
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION
C20	A/R	242	THREADLOCKER, LOCTITE
C21	A/R	2216 B/A	2216 B/A GRAY EPOXY 3M SCOTCH WELD EPOXY
C22	1	2821A001	CANON 52MM 500D CLOSE-UP LENS
C23	1	SDG32104806-008	DECAL, ELECTROSTATIC DISCHARGE IVA 2.88 IN X 1.25 IN, BRADY
C24	2	103MG5	MAGNET, MICRO SWITCH





ISOMETRIC VIEWS
FOR REFERENCE ONLY
SCALE 4/5



(COVER, COVER HARDWARE & HARNESS WIRES REMOVED FOR CLARITY)

NOTES:

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. REMOVED
4. TO SAFELY SECURE CABLE IN STRAIN RELIEF, INSTALL HEAT SHRINK TUBING (ITEM C19) ONTO THE STRAIN RELIEF SADDLES (MAX 2 LAYERS).
5. APPLY THREAD LOCKING COMPOUND (C17) PER ESTABLISHED PROCEDURES.
6. INSERT DRIVE GEAR ASSEMBLY ONTO MOTOR SHAFT UNTIL GEAR CONTACTS SHOULDER, ALIGNING SETSCREW WITH FLAT ON MOTOR SHAFT. MAKE SURE MOTOR TURNS FREE AND CLEAR.
7. DRIVE GEAR AND MATING RING GEAR SHOULD BE ALIGNED CENTER TO CENTER AS CLOSE AS POSSIBLE.
8. TORQUE VALUES SHOWN ARE EFFECTIVE TORQUES ONLY. RUNNING TORQUE MUST BE MEASURED AND ADDED TO THE EFFECTIVE TORQUE VALUES TO DETERMINE THE TOTAL TORQUE.
9. CHEMICAL CONVERSION COAT ANY DAMAGED OR MACHINED AREAS ON ALUMINUM PARTS PER MIL-DTL-5541, TYPE I, CLASS 3.
10. CLEAN ALL SURFACES WITH ISOPROPYL ALCOHOL AND LET AIR DRY. WIPE WITH CLEAN COTTON SWABS AND/OR LINT-FREE CLOTH.
11. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
12. BEFORE STORAGE, PROTECT WITH CONTAMINATION BARRIER. USE CLEAN, NEW, ESD BAG, TOTES OR APPROVED ESD PACKAGING. MARK PART NUMBER, A DASH, AND A SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/SOW. BY TAGGING PER SAE-AS478-35D.
13. CONFIGURE AND TEST PER ACME-TST-019.
14. STAKE THREADED COMPONENTS AS INDICATED WITH ITEM C18, RTV AFTER TESTING AND CONFIGURATION IS COMPLETE.
15. ASSEMBLE PER APPROVED PROCESS PLAN FOR INITIAL BUILD.
16. HANDLE AND CLEAN OPTICAL COMPONENTS, AS NECESSARY, PER ESTABLISHED PROCEDURES.
17. DISCARD SUPPLIED WASHERS AND NUTS FOR ITEM C7.
18. BUILD IN CONJUNCTION WITH S1101EFA2214.
19. PREPARE ITEM C4 ZOOM LENS FOR ASSEMBLY PER ACME-PRO-003. DISCARD COMPONENTS OF ZOOM LENS AS INDICATED IN THE PROCEDURE. REMOVE AND DISCARD INCLUDED CLOSE-UP LENS.
20. APPLY LABEL TO CLEAN, DRY SURFACE IN LOCATION SHOWN.

CHG	ZONE	DESCRIPTION	APP/DATE
G	-	INCORPORATED ACME-ECO-470, ACME-ECO-510 & ACME-ECO-511	AD 12/21/15 JD 12/23/15 KR 12/17/15
F	-	INCORPORATED ACME-ECO-416	AD 03/20/15 JD 03/20/15 GF 03/20/15
E	-	INCORPORATED ACME-ECO-220 & ACME-ECO-350	KR 07/28/14 GF 08/06/14 AD 07/28/14
D	-	INCORPORATED ACME-ECO-205.	MA 09/04/13 KR 09/04/13 AD 09/04/13
C	-	INCORPORATED ACME-ECO-171; INCORPORATED ACME-ECO-177.	MA 07/19/13 JB 07/31/13 AD 07/22/13
B	-	SEE SHEET 2.	L. FLOOR 05/03/13 M. AYNIK 05/07/13 A. DRAKE 05/07/13
	B8 E8 F1	INCORPORATED ACME-ECO-147.	
A	-	SEE SHEET 2.	M. AYNIK, 11/19/12 A. DRAKE, 11/19/12 M. PREVITE, 11/20/12.
	E8	INCORPORATED ACME-ECO-075.	

REVISIONS

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT APPROVED ECO

APPROVED: E.McCLOSKEY DATE: 12/23/15

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

TOLERANCE ON		
.X	± 0.03	PROJECT LEAD: B.BOROWSKI 6/4/12
.XX	± 0.01	QUALITY: D.SEELEY 6/4/12
.XXX	± 0.005	SAFETY: W.BORRELLI 6/4/12
FRACTION	± -	ENGINEER: A.DRAKE 6/1/12
ANGULAR	± 0.50°	DESIGNER: J.CHANGLE 5/10/12
BREAK EDGES	N/A	REVIEWER: M.PREVITE 6/1/12

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
440 625-2200

CONTRACT NUMBER: NNC09BA02B
DELIVERY ORDER: SPACE 110
MODEL NAME: S1101MFA2200

PRIOR TO USE CHECK FOR ANY OUTSTANDING ECO'S WITH CONFIGURATION MANAGEMENT OFFICE

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

ACME DATA CAMERA PACKAGE

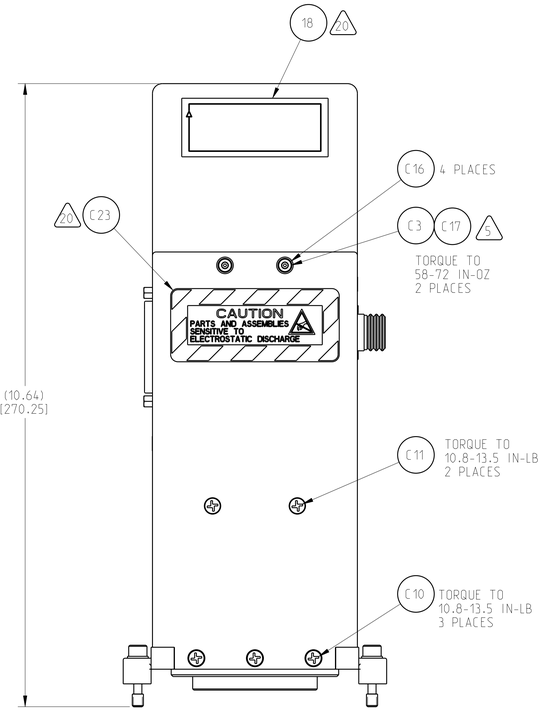
S1101MFA2000

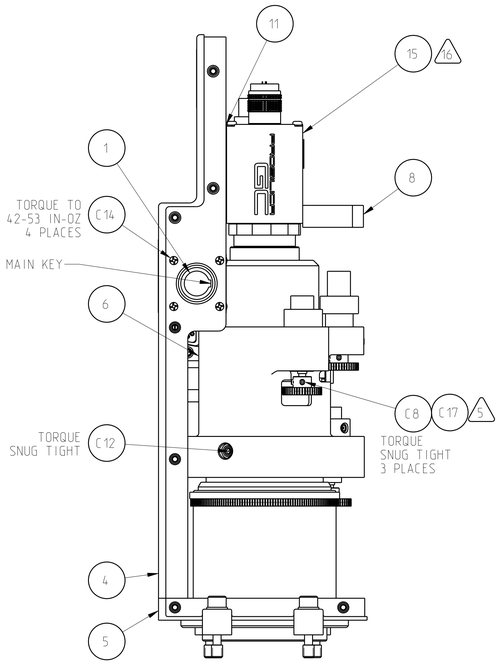
CAMERA ENCLOSURE

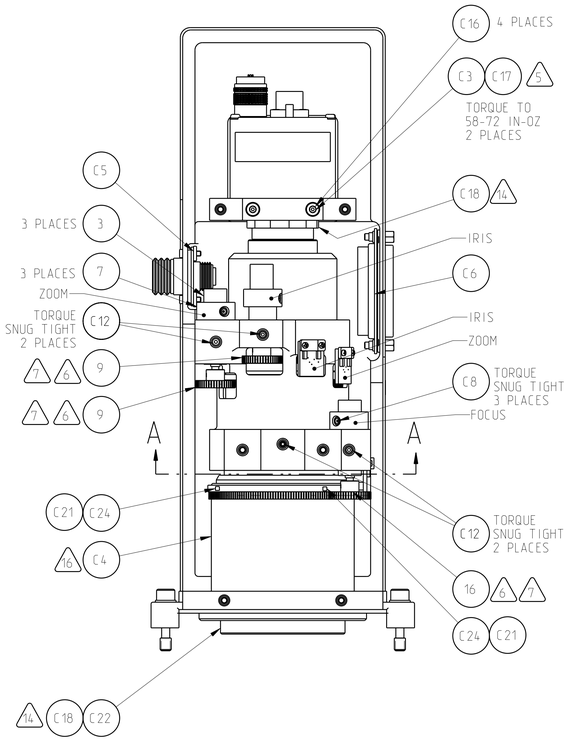
SIZE	DRAWING NUMBER	REV	BLDG	AREA
E	S1101MFA2200	G	ZIN	GRC
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SCALE: 1/1 & NOTED	INITIAL REL DATE: 6/4/12	SHEET: 1 OF 2	DISCIPLINE: MECHANICAL	

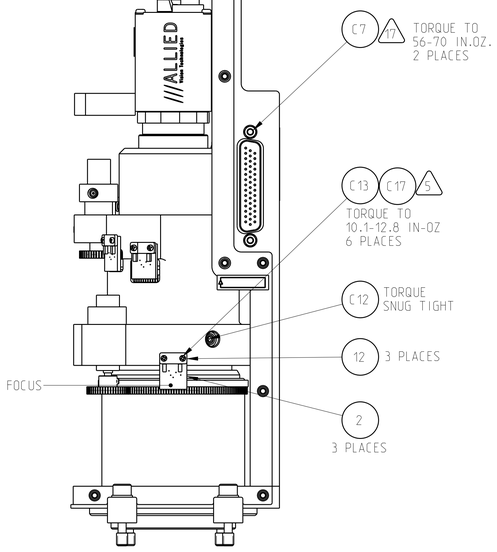
PRO/E VERSION: Wildfire 2.0

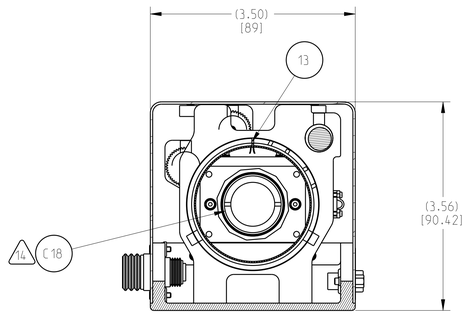
DRAWING NUMBER S1101MFA2200 REV G SIZE E











SECTION A-A
(LENS (C4) REMOVED FOR CLARITY)

(COVER, LENS COVER, COVER HARDWARE & HARNESS WIRES OMITTED FOR CLARITY)

CHG	ZONE	DESCRIPTION	APP/DATE
G	-	SEE SHEET 1.	AD 12/21/15 JO 12/23/15 KR 12/17/15
F	-	SEE SHEET 1.	AD 03/20/15 JO 03/20/15 GF 03/20/15
E	-	SEE SHEET 1.	KR 07/28/14 GF 08/06/14 AD 07/28/14
D	-	SEE SHEET 1.	MA 09/04/13 KR 09/04/13 AD 09/04/13
C	-	SEE SHEET 1.	MA 07/19/13 JB 07/31/13 AD 07/22/13
B	E5 D3,4 B4,5	BALLOON 15 WAS C1 ADDED BALLOON C18 AND FLAG 14 ADDED BALLOON C18 AND FLAG 14	L. FLOOR 05/03/13 M. AYNIK 05/07/13 A. DRAKE 05/07/13
A	E7	BALLOON C3 WAS C2.	M. AYNIK. 11/19/12 A. DRAKE. 11/19/12 M. PREVITE. 11/20/12.

REVISIONS

SIZE	DRAWING NUMBER	REV	BLDG	AREA
E	S1101MFA2200	G	ZIN	GRC
			UNITS: INCH	
SCALE: 1/1	INITIAL REL DATE 6/4/12	SHEET 2 OF 2	DISCIPLINE MECHANICAL	

PRO/E VERSION: Wildfire 2.0

REV G
SIZE E DRAWING NUMBER S1101MFA2200